

Many-analysts simulation study to identify conditions under which results in convenience samples generalize to the target population

1 Background information on the research question to be addressed

Most psychological research relies on **convenience samples**, e.g. student samples, online opt-in panels, clinical samples, self-selected volunteers (Peterson and Merunka, 2014; Andrade, 2021; Wild et al., 2022). This raises a central question: when do results from non-representative samples generalize to the target population? Despite their ubiquity, convenience samples come with little guidance on the conditions under which their results generalize to the target population. This gap is problematic, as convenience sampling can undermine the replicability and generalizability of research findings.

The aim of this project is to assess the degree to which non-representative sampling, operationalized as either (1) restriction in the predictor X or (2) restriction in the outcome Y , affects the following common effect size measures: the unstandardized and standardized regression coefficients (β_1 , β_1^{std}), the Pearson correlation coefficient (r), Cohen’s d (d), and the coefficient of determination (R^2). To this end, we evaluate two hypotheses, each corresponding to one type of restriction and a third hypothesis corresponding to power. Assuming a correctly specified multiple linear model, we formulate the following three hypotheses:

Hypothesis 1: When the sampling mechanism restricts the range or distribution of the predictor X , the unstandardized regression coefficient β_1 and Cohen’s d remain unbiased, whereas β_1^{std} , R^2 and r are in general biased.

Example: If one relies on a sample of university students to estimate the association between intelligence and creativity, the range of intelligence as predictor may be changed because on average university students tend to score higher on intelligence tests than the general population.

Hypothesis 2:

When the sampling mechanism restricts the range or distribution of the outcome Y , all five effect size measures are in general biased.

Example: If one relies on a clinical sample to assess the extent to which trauma and stressful life events influence mental health, then the outcome (mental health) is changed in its range, because patients tend to have poorer mental health than the general population.

Hypothesis 3: Restricting the range or distribution of the predictor X or the outcome Y will lead to a loss of power.

Note that “restriction” can arise from either deterministic truncation or probabilistic sampling mechanisms that alter the range or distribution of a variable.

Range restriction does not merely alter the observed data; it can also induce bias in effect-size estimators, meaning that the expected estimate no longer equals the true population effect size. For both types of restriction, those affecting the predictor X and those affecting the outcome Y , we expect the magnitude of the resulting estimator bias to depend on the severity of the restriction, with stronger restriction leading to greater bias. Moreover, the extent of this bias should vary with the underlying strength of the association between X and Y : when the two variables are uncorrelated, range restriction in either X or Y should not systematically bias any of the effect size measures.

2 Instructions

Identifying conditions under which non-representative sampling mechanisms generalize to the target population requires both a nuanced approach and a pluralistic perspective. We therefore invite you to participate in a confirmatory and in an exploratory part of the present many-analysts simulation study.

2.1 Confirmatory part

In the confirmatory part, we deliberately allow analysts some researcher degrees of freedom (RDF) in specifying the **data-generating process (DGP)** while constraining these choices to ensure correct model specification. This controlled variability enables us to examine how different but reasonable analytic decisions in constructing the DGP influence the resulting bias patterns under range restriction, while still allowing the results to be meaningfully aggregated at the end of the study.

Specifically, you are asked to **implement a data-generating process (DGP)** within a pre-defined script of your choice (either .R or .Rmd) that can be used to evaluate Hypotheses 1 -3, under the following constraints:

Model structure: The DGP should include a continuous outcome Y and a predictor of interest X , which may be either continuous or binary. Participants may alternatively provide two separate scripts, one with a continuous X and one with a binary X . The relationship between X and Y should be linear and additive, without interactions or nonlinear terms. Errors should be independent and homoscedastic:

$$\varepsilon \sim \mathcal{N}(0, \sigma^2)$$

A minimal example is:

$$y = \beta_0 + \beta_1 x + \varepsilon$$

You may, however, include additional covariates (e.g., confounders correlated with X) and define X from any distribution, such as normal, uniform, or an empirical one.

Range restriction: To test the hypotheses, your DGP should implement sampling mechanisms that systematically alter the range or distribution of either X or Y . You are free to choose any mechanism that produces different levels of restriction, such as:

- excluding observations above or below a specific threshold or percentile (deterministic);
- probabilistic sampling, where the probability of inclusion depends on the restricted variable - on X when the restriction is applied to X , and on Y when the restriction is applied to Y (probabilistic).

Function specification: Your DGP function must be named `dgp()` and include three mandatory arguments:

1. **beta1:** defining three scenarios concerning the strength of association between X and Y (weak, moderate, strong) by varying β_1 while keeping all remaining parameters fixed. Note that all values for **beta1** need to be different from zero.
2. **restrict:** specifying the level of range restriction and can take one of four character values: **none**, **weak**, **moderate**, or **strong**. These levels determine how extreme the resulting convenience samples are by implementing sampling mechanisms that induce weak, moderate or strong restriction in both X and Y . The corresponding parameters that control the strength of restriction should be fixed within the DGP function for each level. Note that the four levels are defined **within** each variable, i.e., “strong restriction” in X is not necessarily directly comparable to “strong restriction” in Y , as the marginal distributions of X and Y may differ.
3. **n:** the number of observations in each sample, which may be chosen freely. For comparability, the unrestricted sample and the samples restricted on X and Y must all contain the same number of observations n .

Output: Your `dgp()` function should return a list with two data frames:

- `convenience_x`: sample restricted on `X`;
- `convenience_y`: sample restricted on `Y`;

If `restrict = "none"`, i.e., when the argument `"none"` is passed to `restrict`, no restriction is applied. In this case, `convenience_x` and `convenience_y` should both contain the same unrestricted dataset. Each data frame should contain at least a continuous or binary predictor `X`, and a continuous outcome `Y`. You may add other covariates if desired.

Optional arguments: You may include additional arguments with fixed default values in your `dgp()` function, e.g., `dgp(beta1, restrict, n, sigma = 10)`. These defaults must comply with the confirmatory design but may be changed later in the exploratory part.

Model formula: After defining your `dgp()` function, please define an R model formula (e.g., `y ~ x + cov1 + cov2`). The formula should include `X` and any covariates that are correlated with both `X` and `Y`.

2.2 Exploratory part (optional)

In addition to the confirmatory analyses, you are invited to extend your simulation with an exploratory part. This allows you to investigate additional conditions or variations that provide a broader perspective on how convenience samples generalize to the target population.

You may, for example, explore:

- violations of model assumptions (e.g., correlated or heteroscedastic errors),
- measurement error in the predictor `X`,
- no effect ($\beta_1 = 0$)
- different sample sizes, distributions of `X`, or sampling mechanisms.

Such extensions may reveal when even the unstandardized regression coefficient becomes biased or when bias increases under more severe range restriction.

You may add your exploratory analyses under the *Exploratory part* section at the end of the script and document them with a short written description.

3 Implementation

Setup

Before running the script of your choice (either a `.R` or a `.Rmd` script) check that you defined the data generating function `dgp(beta1, restrict, n)`, the parameter values for `beta1` and `n`, and the model formula and received the `Helper.R` script. This script incorporates multiple validation checks for the confirmatory study to verify that all restrictions are met and that every component is correctly specified. It additionally provides a fixed seed for the simulations and contains the complete code used to perform the simulations and compute the effect size estimates, relative bias and power. Also code to represent the results is provided in this script.

You are encouraged to include comments in your code, especially in the exploratory section.

If the script does not run and there are no error messages check again the upper rules for the `dgp(beta1, restrict, n)` function.

If you do not find the problem, feel free to contact us: multi.analyst@stat.uni-muenchen.de

```
# load helper functions into the current R session
# make sure the file is in you current working direction
source("Helper.R")
```

DGP

```
dgp <- function(beta1, restrict = c("none", "weak", "moderate", "strong"), n, ...){
  # Template: implement the data-generator and return two data.frames with columns:
  #   - x : numeric or binary predictor
  #   - y : numeric outcome
  # Additional variables (covariates) may be returned as extra columns.
  #
  # define the DGP

  # no restriction
  # if(restrict == "none"){
  #   convenience_x <- convenience_y <- ...
  # }

  # restrict by x

  # define weak restriction:
  # if(restrict == "weak"){
  #   ...
  # }

  # define moderate restriction:
  # if(restrict == "moderate"){
  #   ...
  # }

  # define strong restriction:
  # if(restrict == "strong"){
  #   ...
  # }

  # restrict by y

  # define weak restriction:
  # if(restrict == "weak"){
  #   ...
  # }

  # define moderate restriction:
  # if(restrict == "moderate"){
  #   ...
  # }

  # define strong restriction:
  # if(restrict == "strong"){
  #   ...
  # }
```

```
# }

# list(convenience_x = , convenience_y =)
# keep this and include your data.frame behind the =
}
```

Effect size

```
beta1 <- c(NA, NA, NA)
```

Number of observations per sample

```
n <- NA
```

Model formula

```
# formula <- y ~ x + cov1 + cov2
# your data.frames from dgp should include these variables!
```

4 Results of the Monte Carlo design

This section repeats your data-generating procedure for n_{iter} Monte Carlo iterations (defined as `niter`; by default this is set to 10,000, although we recommend using 100 for testing purposes). This will estimate the variability and bias of your effect size measures across repeated samples. To obtain the results, simply run the sections below.

Run the checks: Before running the full simulation, execute the `checks()` function. If no errors or warnings occur, your DGP is valid and you can proceed with the simulation.

Run the simulation: Once the checks pass, use the `run_simulations()` function to perform the Monte Carlo iterations.

The wrapper collects per-iteration results and then computes summary quantities across iterations (mean relative bias, Monte Carlo standard error of the mean, and 95% simulation interval).

```
# suggested number for reliable results niter = 10000
niter <- 100 # number of iterations for test purposes

# run checks before running the simulation
# if no error occurs you can run the simulation sim
checks(niter, beta1, restrict, n, formula, dgp)

# run if you got the go from the checks()-function
sim <- run_simulations(niter, beta1, restrict, n, formula, dgp)
```

Represent the results: Run the two following functions. `simulation_results()` returns two figures showing the results for sample restriction of **X** and **Y**, respectively. Note: the Pearson correlation coefficient is only derived for a numeric **X** and the cohen's d only for a binary **X**.

The function `power_loss_table()` returns a figure displaying the power for the unrestricted sample and the convenience samples.

All figures will be saved in the same folder where your `Helper.R` script is located. Please send us your completed script along with the three resulting figures and the `sessionInfo` text file generated at the end of your script.

```
simulation_results(sim, restrict)
# loss in power
power(sim)
```

Exploratory part

In the exploratory part, you are free to extend the confirmatory simulations with any settings you consider relevant or interesting. There are no restrictions on the DGP or the types of violations you introduce (e.g., model misspecification, nonlinearity, measurement error, etc.). Please document your implementation and design choices in a short written report.

You may, however, reuse the functions in `Helper.R` to the extent that your exploratory design remains compatible with their assumptions.

When you can use `run_simulations()`: The function `run_simulations()` can be used as long as the following arguments are specified in the required format:

- `niter`: an integer giving the number of Monte Carlo iterations.
- `beta1`: a numeric vector of length three, specifying the true effect sizes to be evaluated.
- `restrict`: a character vector containing the four restriction levels "none", "weak", "moderate", and "strong".
- `n`: an integer specifying the sample size per simulated dataset.
- `formula`: an R formula specifying the analysis model (e.g., $y \sim x + z$).
- `dgp_fun`: a data-generating function of the form `dgp_fun(beta1, restrict, n)` as described above, which returns the required samples (unrestricted, restricted on X, restricted on Y) in the same structure as in the confirmatory part.

Within this framework, you may modify the DGP quite freely, for example, by introducing heteroscedastic errors, nonlinear terms, effect modification, or measurement error, provided that the arguments and output structure expected by `run_simulations()` and `one_simulation_run()` are preserved.

When the Helper functions are no longer sufficient: The current Helper functions are tailored to the effect size measures used in the confirmatory part (unstandardized and standardized regression coefficients, Pearson's r , Cohen's d , R^2) and the corresponding confidence intervals and power. If, in your exploratory analyses, you wish to:

- derive additional effect size measures (e.g., Spearman rank correlation, odds ratios, partial correlations), or
- focus on other performance metrics (e.g., predictive performance, calibration measures, classification accuracy),

then you will need to implement the computation of these quantities yourself (e.g., by extending `model_function()`, adding new summary code, or writing separate analysis functions). In such cases, `run_simulations()` and the plotting functions (`simulation_results()`, `power()`) may serve as templates, but will not automatically handle the new measures without modification.

5 Reproducibility information (required):

Please run the following code at the end of your script. This will generate a .txt file containing your R version and package versions. Please submit this file together with your script and outputs.

```
writeLines(  
  capture.output(sessionInfo()),  
  paste0("sessionInfo", Sys.Date(), ".txt")  
)
```

References

- Andrade, C. (2021). The inconvenient truth about convenience and purposive samples. *Indian Journal of Psychological Medicine*, 43(1):86–88. PMID: 34349313.
- Peterson, R. A. and Merunka, D. R. (2014). Convenience samples of college students and research reproducibility. *Journal of Business Research*, 67(5):1035–1041.
- Wild, H., Kyröläinen, A.-J., and Kuperman, V. (2022). How representative are student convenience samples? A study of literacy and numeracy skills in 32 countries. *PLOS ONE*, 17(7). Publisher: Public Library of Science.